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Global MPPT for PV Systems under Partial Shading Conditions using PSO with Tuned PI Controller

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1. INTRODUCTION

1.1 Background

The global energy sector is undergoing a rapid transition towards decarbonization, driven by the urgency of climate change mitigation and the falling cost of renewable generation technologies. Among the available renewable options, solar photovoltaic (PV) technology has become particularly attractive because of its modularity, low maintenance requirement, and the steep decline in its levelized cost of electricity over the last decade. Solar module prices have fallen by roughly 20% for every doubling of global installed capacity [1].

Nepal's electricity system is presently dominated by hydropower, which exposes the grid to strong seasonal variability, since river flows and hence generation fall sharply in the dry winter months. Recent assessments indicate that Nepal possesses a very large untapped solar resource, with average insolation of about 4.7 kWh/m²/day and roughly 300 sunshine days a year, while installed solar capacity remains a small fraction of the country's hydropower capacity [2].

A PV module converts incident solar radiation into electrical energy through the photovoltaic effect, but its current-voltage (I-V) and power-voltage (P-V) characteristics are highly nonlinear and depend strongly on solar irradiance and cell temperature. Under uniform irradiance, each operating condition has a single point on the P-V curve, called the Maximum Power Point (MPP), at which the array delivers its maximum available power. Because the array's internal impedance rarely matches the load impedance at this point, a power-electronic interface (typically a DC-DC converter) together with a Maximum Power Point Tracking (MPPT) controller is required to continuously adjust the array's operating voltage so that it tracks the MPP as irradiance, temperature, and load conditions change.

A wide range of MPPT algorithms exist, from simple hill-climbing methods such as Perturb and Observe (P&O) and Incremental Conductance (IC), to more elaborate soft-computing and advanced techniques. Conventional hill-climbing algorithms are simple to implement and perform adequately under uniform irradiance, but, as discussed in Section 1.2, they have well-documented limitations [3] both in their transient/steady-state trade-off and in their inability to correctly locate the true maximum power point once the PV array is partially shaded.

1.2 Problem Statement

Conventional hill-climbing MPPT algorithms such as P&O periodically perturb the array's operating voltage (or the converter duty cycle) and observe the resulting change in power to decide the direction of the next perturbation. The size of this perturbation step directly governs performance: a large step size gives a fast dynamic response when irradiance changes suddenly, but causes the operating point to oscillate around the MPP in steady state, wasting energy; conversely, a small step size reduces this steady-state oscillation but slows the algorithm's response

to changing conditions. This inherent trade-off means that a bare P&O algorithm cannot simultaneously be fast and low-ripple, motivating the introduction of an auxiliary closed-loop PI controller that regulates the converter duty cycle towards the voltage reference generated by the search algorithm, decoupling the tracking decision from the power-stage dynamics and allowing both fast response and low oscillation to be achieved together.

A more fundamental problem arises under Partial Shading Conditions (PSC), which occur routinely in practical rooftop and ground-mounted installations because of moving clouds, dust accumulation, bird droppings, or shadows cast by nearby trees, poles, and buildings. When series-connected PV modules or cells receive unequal irradiance, the associated bypass diodes activate to prevent hotspot damage, and the array's P-V curve develops multiple local power peaks in addition to a single global peak, the Global Maximum Power Point (GMPP). Because conventional hill-climbing algorithms only compare the present operating point with the immediately preceding one, they cannot distinguish a local peak from the true global maximum, and they typically settle on whichever peak lies closest to their starting point. This can strand the system at a fraction of the power that is genuinely available, even though the PI-controller tuning problem described above may already have been resolved. A global search technique, one capable of exploring the entire operating range rather than climbing along a single local gradient, is therefore required to reliably locate the true GMPP under PSC, while a well-tuned PI controller is still needed for the fine, low-ripple regulation once the correct region has been found.

1.3 Objectives

- To model a PV array interfaced with a DC-DC converter in MATLAB/Simulink and characterize its P-V behavior under varying irradiance (at a fixed cell temperature of 25 °C).
- To implement the conventional Perturb and Observe MPPT algorithm with a PI controller, and evaluate its tracking performance under uniform and step-varying irradiance.
- To simulate the PV array under partial shading conditions and demonstrate the failure of the conventional algorithm to converge on the global maximum power point.
- To design a Particle Swarm Optimization (PSO) based global MPPT algorithm and evaluate its ability to converge to the true global maximum power point under partial shading.
- To comparatively analyze the behavior of the P&O scheme against the proposed PSO-based global scheme.

1.4 Scope and Limitations

The scope of this project is restricted to a simulation-based study carried out entirely in MATLAB/Simulink. The PV array is modeled feeding a DC-DC buck converter connected to a constant resistive load, and only DC-side maximum power extraction is considered.

The load is assumed constant rather than dynamically varying, and grid-connection aspects such as a DC-AC inverter stage, synchronization, and power quality are outside the scope of this phase.

Environmental conditions are represented using a set of representative step changes and partial-shading patterns, with the cell temperature held constant at 25 °C, rather than continuously logged real-world weather data. The PSO parameters are tuned through simulation trials for the shading patterns rather than through an exhaustive search across every possible configuration.

This work does not extend to hardware prototyping or experimental validation.

2. LITERATURE REVIEW

Maximum power point tracking (MPPT) has been studied extensively, because an efficient MPPT method is vital to extract maximum power and improve efficiency of PV power generation, which can affect investment cost and cost of electricity. Podder, Roy, and Pota [4] provide a critical review of MPPT methods classified by their tracking nature, distinguishing conventional hill-climbing techniques from intelligent, optimization-based and hybrid approaches, and observe that the appropriate choice of technique depends strongly on the operating scenario a system is expected to face, particularly whether uniform or non-uniform irradiance is anticipated.

Pairing a hill-climbing search algorithm with a dedicated closed-loop controller improves tracking quality compared with driving the converter switch directly from the search algorithm. de Brito et al. [5] show that recasting the MPPT search as a control-system design problem, with explicit voltage and current control loops, yields a markedly higher tracking factor than open-loop duty-cycle perturbation. In the same context, Harrison et al. [6] implemented a PI-controlled P&O scheme in MATLAB/Simulink driven by measured irradiance and temperature data logged from an experimental PV panel, and found that introducing the PI controller considerably reduced the power spikes and oscillation observed when the converter was driven by P&O alone, so that the power delivered to the load followed the available input power far more closely. The control structures in [5] and [6] are very different: [5] uses a more complex arrangement of cascaded voltage and current loops, whereas [6] uses a single PI loop fed by the MPP voltage reference and the PV output voltage. The simpler single-loop structure is adopted in this project (Figure 3.2). More recently, Deshi et al. [7] combined an enhanced P&O algorithm with PI gains tuned by accelerated particle-swarm and adaptive spiral-dynamic optimization on high-power boost converters, reporting sub-second settling times, negligible overshoot, and extraction efficiencies above 98%; their results reinforce that the quality of PI tuning, and not merely the presence of a PI stage, has a direct bearing on the achievable transient and steady-state performance.

Wasim et al. [8] and Nagadurga, Devarapalli, and Knypiński [9] both explain how partial shading, through the activation of module bypass diodes, produces a P-V curve with several local power peaks alongside a single global peak, and that gradient-following algorithms such as P&O are structurally unable to tell a local peak from the global one.

Wirateruna and Millenia [10] applied Particle Swarm Optimization directly to GMPPT under partial shading and reported that the swarm-based search reliably located the true global peak where conventional methods failed. Their results indicate that PSO can handle partial shading with

a fast response, which makes it a suitable candidate for tracking the optimum operating point under partial shading conditions.

The literature review points to two main findings: first, that a PI-controlled voltage loop improves the transient and steady-state quality of hill-climbing MPPT under uniform irradiance; and second, that such hill-climbing methods cannot reliably find the global peak once the array is partially shaded, a task for which swarm-based search such as PSO is far better suited. This project therefore studies both approaches within the same simulation framework so that they can be compared directly.

Thus, this Bachelor's-level minor project provides a detailed study of the P&O algorithm with a tuned PI controller under uniform and varying irradiance, demonstrates its failure under PSC, and presents PSO as a solution for PSC.

3. METHODOLOGY

The flowchart in Figure 3.1 outlines the methodology and tasks for this minor project.

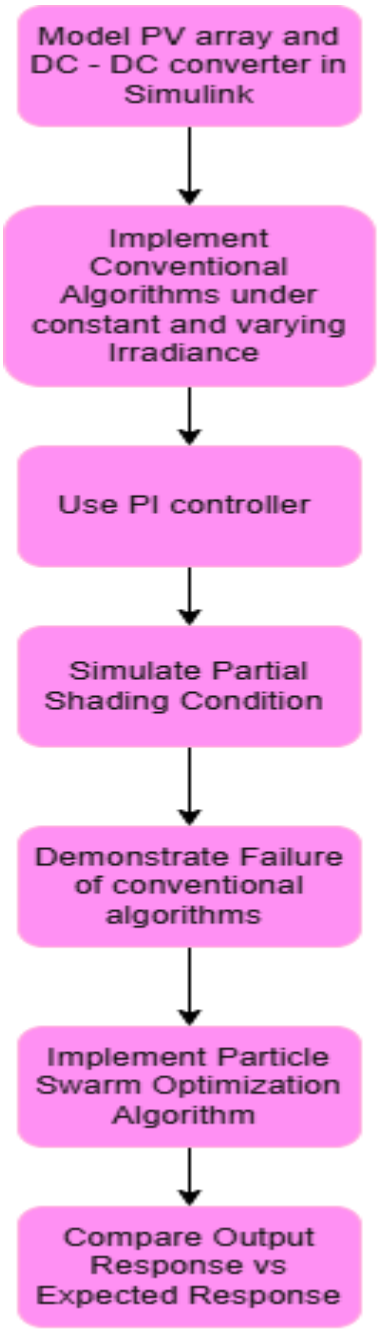


Figure 3.1: Flowchart of the Methodology.

The project is carried out through simulation in MATLAB/Simulink, following the sequence of steps described below. Figure 3.2 shows the overall closed-loop system that was built in Simulink. The MPPT algorithm (P&O or PSO) uses the measured PV voltage and current to generate a voltage reference V_ref, which is compared with the measured PV voltage V_PV; the resulting

error is processed by a PI controller whose output drives the PWM generator and hence the DC-DC converter.

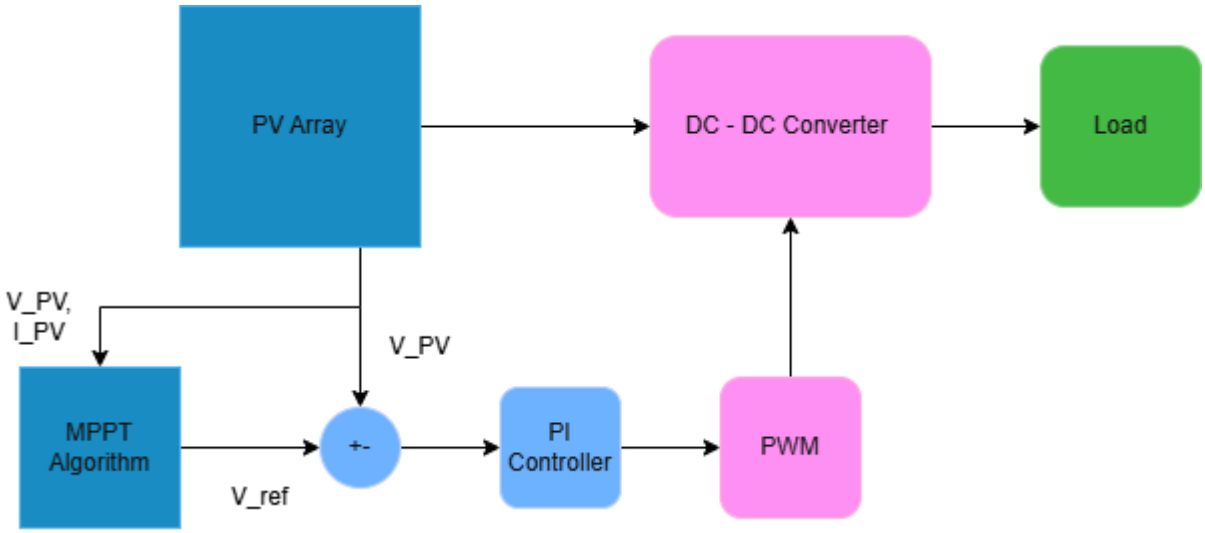


Figure 3.2: Closed-loop PV – MPPT – Buck converter system modeled in Simulink

3.1 PV Array and Buck Converter Modeling

A PV array is modeled in Simulink using the built-in PV Array block (Figure 3.3), configured as 2 series modules in 1 parallel string. The resulting I-V and P-V characteristics are plotted in Figure 3.4. Under standard conditions (1000 W/m², 25 °C) the array has an open-circuit voltage of about 72 V, a short-circuit current of about 7.9 A and a maximum power of about 425 W at around 58 V (Figure 4.1). The cell temperature is kept at 25 °C throughout the study, so irradiance is the only varied environmental input.

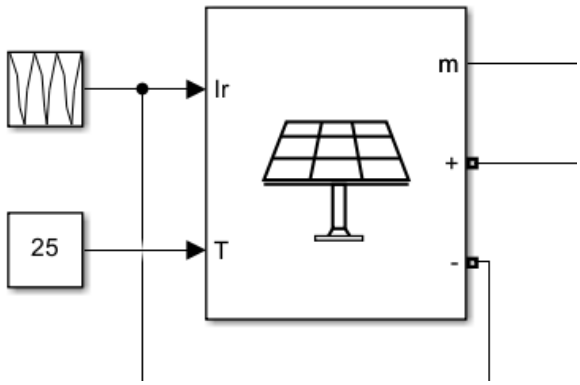


Figure 3.3: PV array in-built model in Simulink

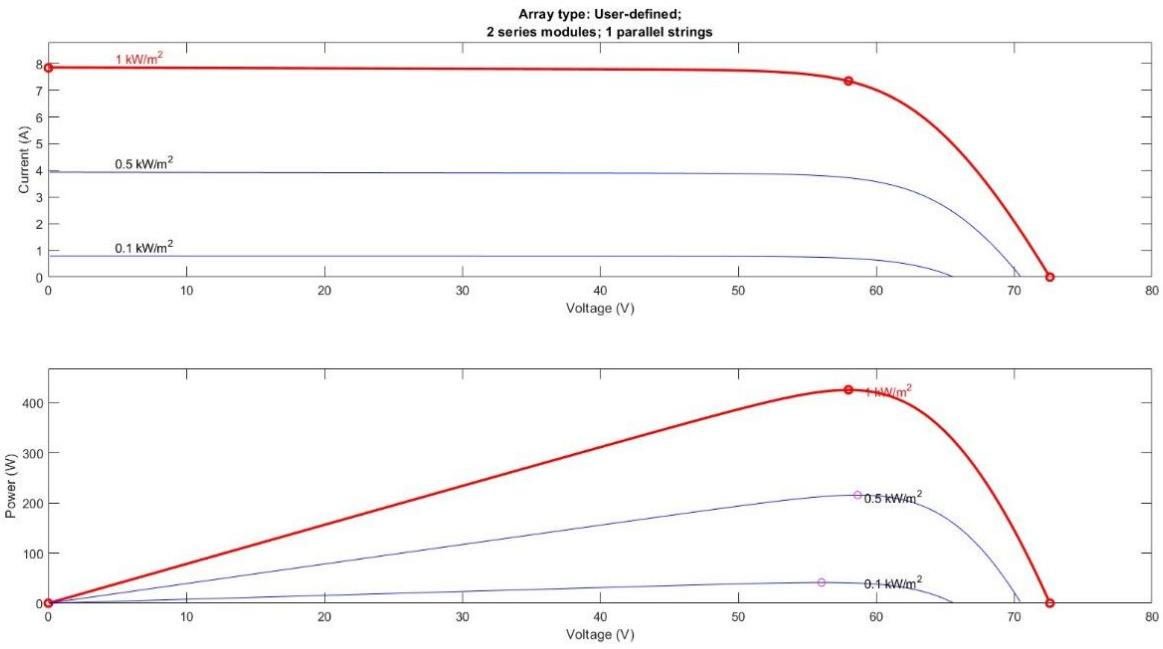


Figure 3.4: I-V and P-V characteristics of 2 series, 1 parallel PV array.

A DC-DC buck converter, consisting of an inductor, a switch (MOSFET), a diode, input and output capacitors, and a resistive load, is connected to the PV array output (Figure 3.5). To avoid rigorous design calculations, conservatively large component values were chosen: both the input and the output capacitor are 1000 μF, and the inductor is 100 mH, which keeps the voltage and current ripple small.

For simplicity, a constant resistive load of 1 Ω is used.

A step-down stage is suitable here because, with a 1 Ω load, the resistance seen by the array is R/D^2, and the array's MPP resistance (about 8 Ω at 1000 W/m²) can therefore be matched with a duty cycle D of roughly 0.35.

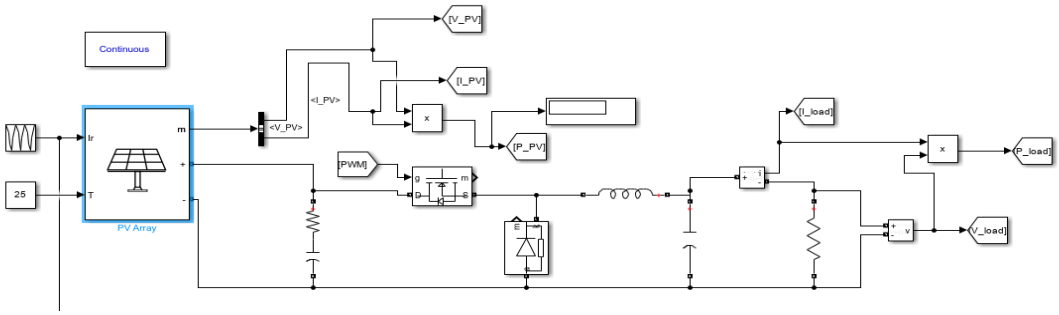


Figure 3.5: PV Array connected to a load through DC-DC Buck Converter

3.2 Conventional MPPT: PI-Tuned Perturb and Observe

The Perturb and Observe algorithm is implemented as shown in Figure 3.6: the array voltage and current are sampled each cycle, the instantaneous power is computed, and the change in power relative to the previous sample determines whether the voltage reference should be increased or decreased (Appendix I). If the power has increased, the perturbation continues in the same direction; otherwise it is reversed. The generated reference voltage, limited to the range 10–70 V, is compared with the actual array voltage to form an error signal, which is regulated by a PI controller whose output sets the buck converter's duty cycle through a PWM generator, exactly as depicted in Figure 3.2. For comparison, P&O is also run without the PI stage, perturbing the converter directly (Section 4.3).

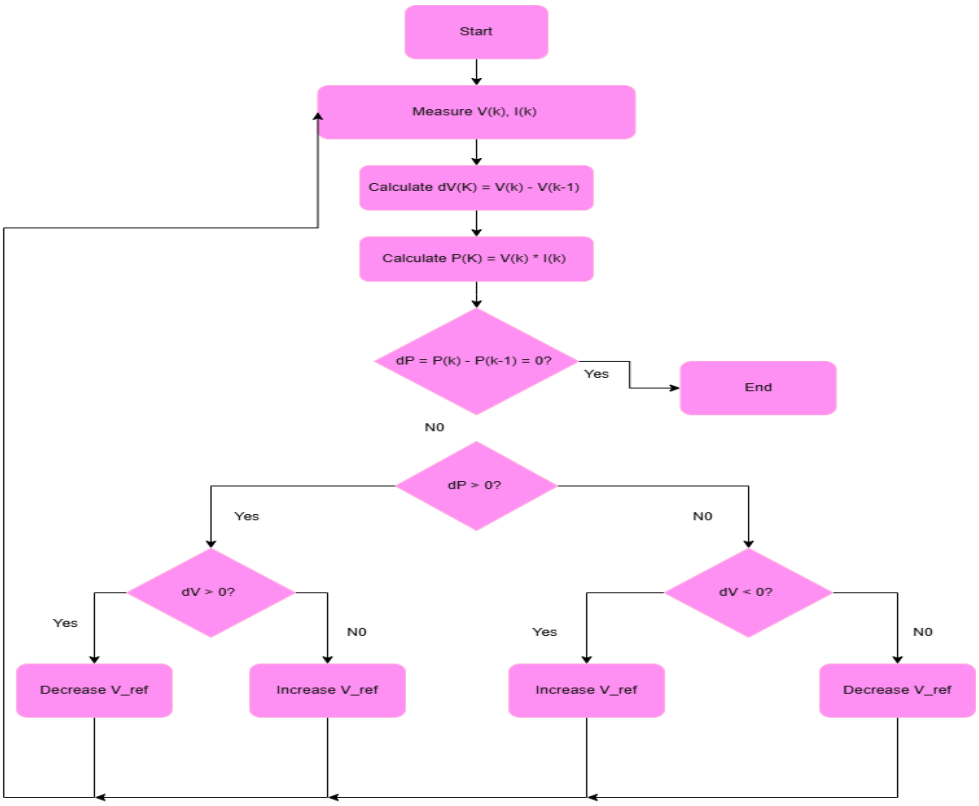


Figure 3.6: Flowchart of the Perturb and Observe (P&O) MPPT algorithm

3.2.1 PI Controller Tuning

The proportional and integral gains must be set carefully, and several methods are available. One option is the Bode-plot method, in which the gains are chosen so that the phase margin lies between 45° and 60°, giving a fast transient response without excessive overshoot or steady-state oscillation. This technique is analytically accurate but requires a reliable small-signal model of the converter and PV source. Another option is the auto-tune function in Simulink, which avoids this effort but may not always give the desired response.

Due to time constraints, the proportional and integral gains were instead set by trial and error: several (Kp, Ki) pairs were simulated and compared on the basis of overshoot, steady-state oscillation and speed of response (Sections 4.4 and 4.8).

3.3 Performance Evaluation under Uniform and Varying Irradiance

With the PI-tuned P&O scheme in place, the system is subjected to step and ramp changes in irradiance at constant temperature (for example, a step from 1000 W/m² to 400 W/m²), and its power tracking response is recorded with Simulink scopes. This establishes a performance baseline against which the partial-shading behavior and the proposed global scheme will later be compared.

3.4 Partial Shading Simulation

Partial shading is simulated by connecting four PV array blocks in series, each with its own bypass diode, and exposing them to different irradiance levels, reproducing the multi-peak P-V characteristic described in Section 1.2 and illustrated in Figure 3.7.

Specifically, two of the four arrays receive a constant irradiance of 1000 W/m², while the other two receive 1000 W/m² for the first part of the simulation and then drop to 400 W/m², so that half of the string is partially shaded (Figure 4.8).

The PI-tuned P&O scheme from Section 3.2 is run on this shaded array, and its final operating point is compared against the true GMPP value in order to quantify the power lost when the conventional algorithm converges on a local peak.

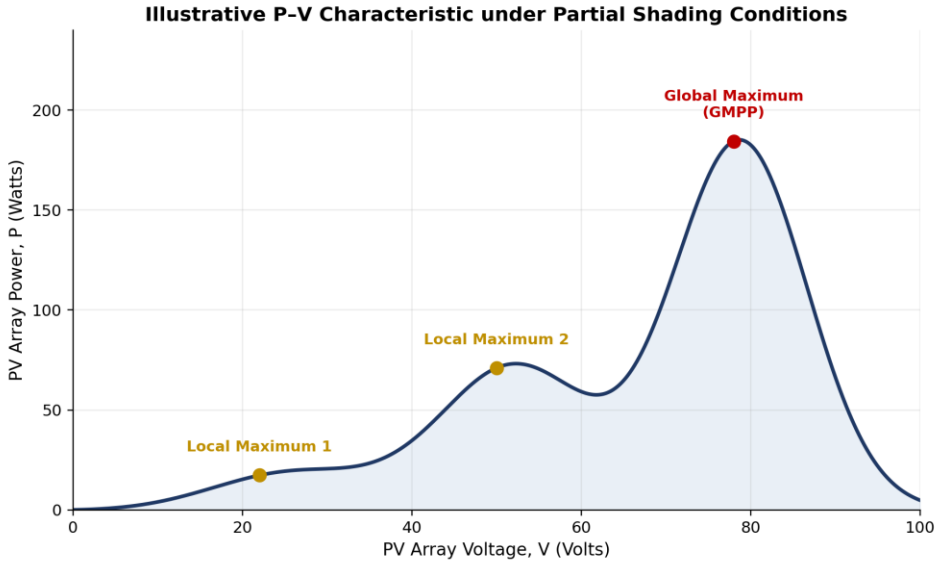


Figure 3.7: Illustrative multi-peak P-V characteristic under partial shading conditions

3.5 Global MPPT Using Particle Swarm Optimization

A Particle Swarm Optimization based global MPPT algorithm is designed, in which each particle represents a candidate array operating voltage, used as the reference voltage V_ref of the PI-controlled converter (equivalently, a candidate converter duty cycle), as shown in Figure 3.8. A swarm of N = 4 particles is initialized at evenly spaced voltage values spanning the search range of 5–70 V, with zero initial velocities.

At each iteration, the array power corresponding to every particle's voltage is measured (the particles are evaluated one after another, one per sampling instant, so that the converter can settle at each candidate point). Each particle's personal best (pbest) is updated whenever it records a higher power than before, and the swarm's global best (gbest) is updated whenever any particle exceeds the best power found by the swarm so far. Once every particle has been evaluated, velocities and positions are updated using the standard PSO equations, which combine an inertia term with cognitive (pbest-directed) and social (gbest-directed) attraction terms.

The velocity and position of particle i at iteration k are updated using the following PSO equations:

$$v_i^{k+1} = w v_i^k + c_1 r_1 (\text{pbest}_i - x_i^k) + c_2 r_2 (\text{gbest} - x_i^k)$$

and the position of each particle is then updated by:

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

Here, w is the inertia weight, set to 0.5, which makes the particle tend to continue moving in its previous direction. The cognitive term, with acceleration coefficient $c_1 = 1.5$ and random number r_1, pulls a particle towards the best position it has itself discovered (pbest). The social term, with coefficient $c_2 = 2.0$ and random number r_2, pulls it towards the best position discovered by the whole swarm (gbest).

The numbers r_1 and r_2 are random values drawn uniformly from [0, 1] at every update. They prevent all particles from moving identically; without randomness, the swarm could become too deterministic and lose some of its exploration capability.

In the implemented controller (Appendix II) the swarm iterates continuously: as the particles converge towards gbest, the reference voltage settles at the voltage of the located GMPP. No explicit stopping criterion or restart on irradiance change is used; a stopping rule (a maximum

number of iterations, or the power variation across the swarm falling below a small threshold) and a restart trigger are possible refinements.

The maximum velocity (step size) is limited to 5 V per iteration to prevent large jumps that may cause large voltage oscillations and instability, and the particle positions are bounded between 5 V and 70 V (close to the array's open-circuit voltage) to prevent moves to extreme values.

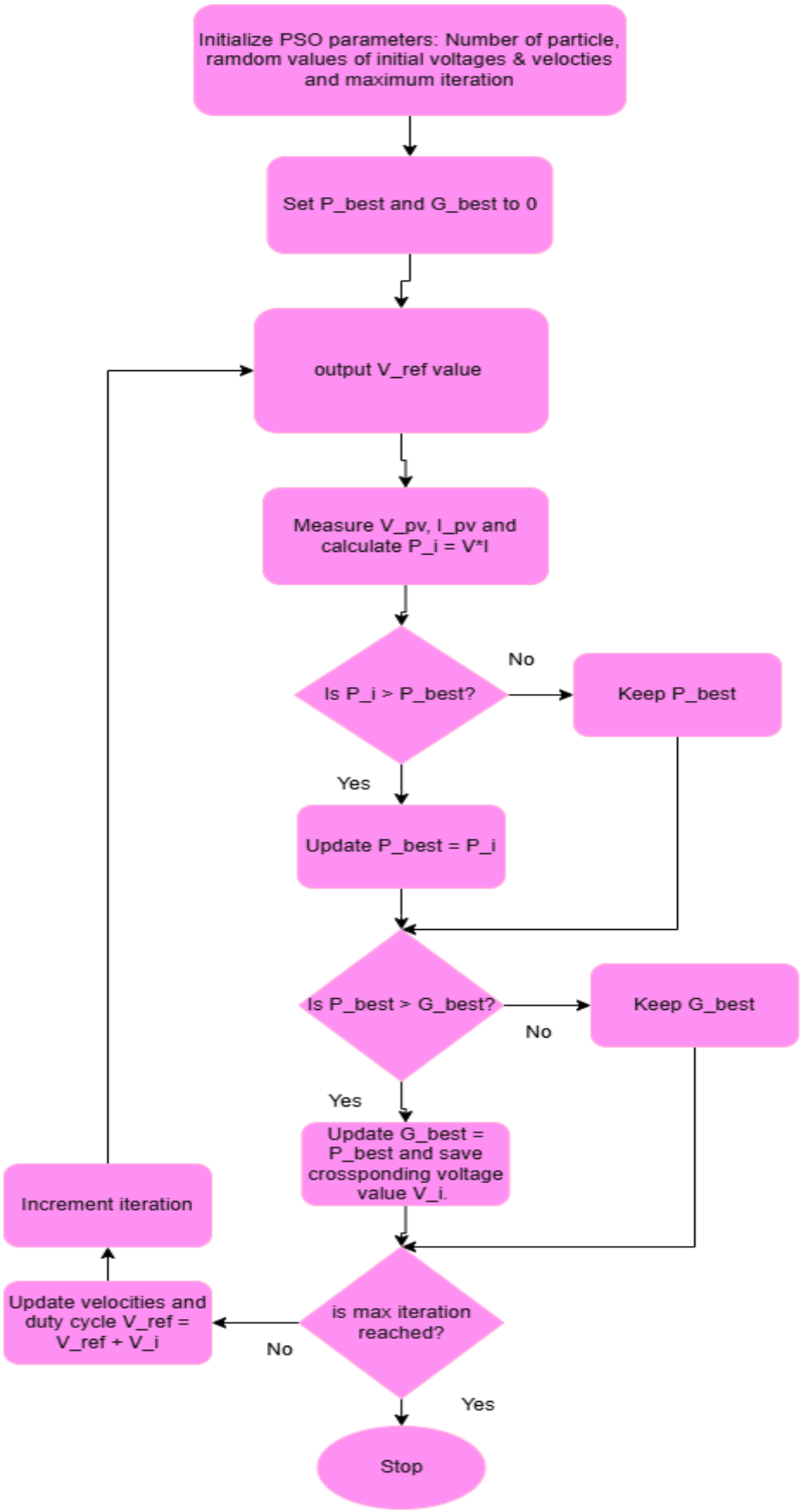

Initialize PSO parameters: Number of particle, ramdom values of initial voltages & velocties and maximum iteration
Set P_best and G_best to 0
output V_ref value
Measure V_pv, I_pv and calculate P_i = V*I
Is P_i > P_best?
No
Keep P_best
Yes
Update P_best = P_i
Is P_best > G_best?
Keep G_best
No
Yes
Update G_best = P_best and save crossponding voltage value V_i.
Increment iteration
Update velocities and duty cycle V_ref = V_ref + V_i
is max iteration reached?
No
Yes
Stop

Figure 3.8: Flowchart of the proposed PSO-based Global MPPT algorithm

3.6 Comparative Performance Analysis of P&O and PSO Algorithm

The conventional PI-tuned P&O scheme and the proposed PSO-based global scheme are evaluated side by side across partial-shading patterns on the basis of output power tracking capability, speed of convergence and steady-state oscillation. These results form the basis for the project's conclusions.

3.7 Scope for Future Expansion

Beyond the scope of this project, the work could be extended by adding a grid-tied DC-AC inverter stage with synchronization, evaluating alternative converter topologies, introducing a dynamically varying load, scaling the model to a larger solar farm with many parallel strings, integrating a battery energy storage system, implementing a hybrid PSO–P&O scheme with a restart trigger on irradiance change, and validating the simulated results through a physical prototype implementation.

4. RESULTS AND DISCUSSION

4.1 PV Characteristics

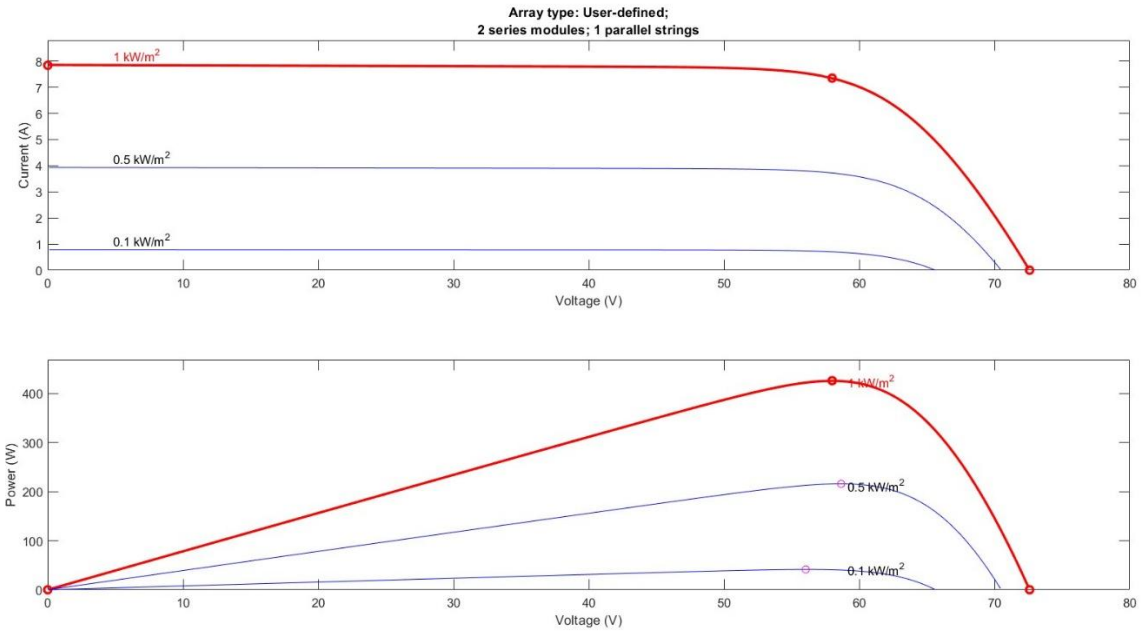


Figure 4.1: PV Characteristics of 2 series and 1 parallel PV Array

For uniform irradiance, the 2-series, 1-parallel PV array (Figure 4.1) delivers a maximum power of about 425 W at just under 60 V (≈ 58 V) at 1000 W/m², with an open-circuit voltage of about 72 V and a short-circuit current of about 7.9 A. At 500 W/m² the maximum power is about 215 W at a similar voltage, and at 100 W/m² it is about 40 W at around 56 V. The maximum power therefore scales almost in proportion to irradiance, while the MPP voltage stays within a narrow band (about 56–59 V), because irradiance mainly changes the current and affects the voltage only slightly. This narrow voltage band is what allows a voltage-reference-based tracker with fixed search limits to work across the whole irradiance range.

4.2 Fixed Duty Cycle Output under Continuously Varying Irradiance

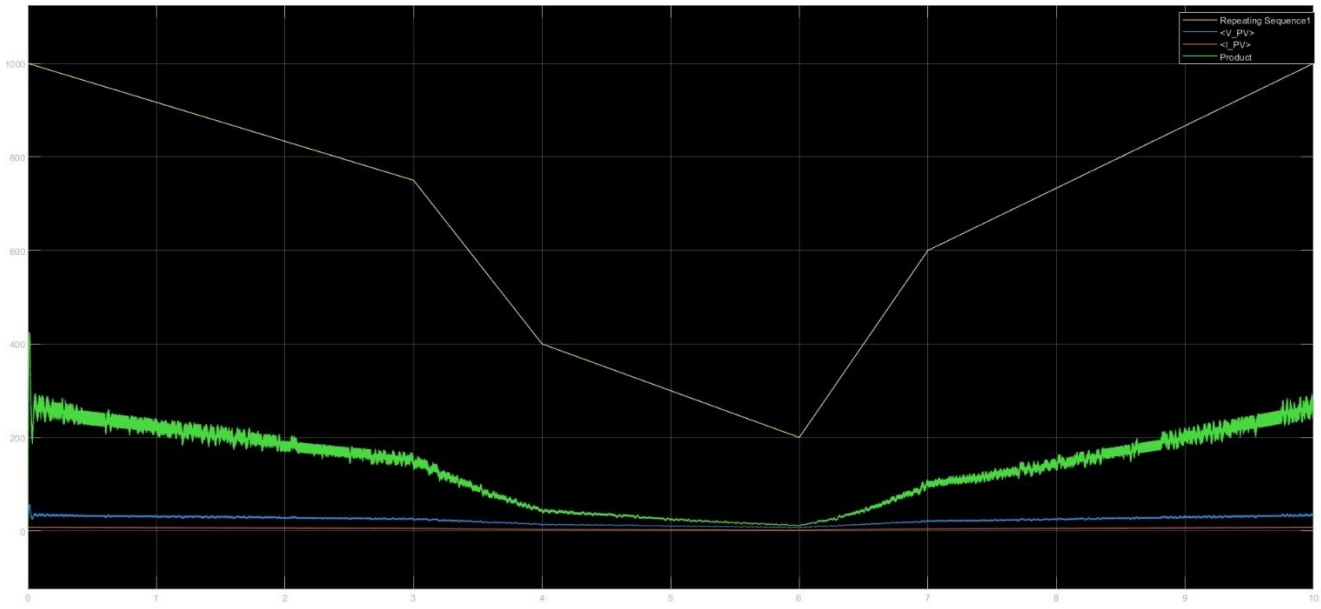


Figure 4.2: Fixed Duty Cycle output under constantly varying irradiance

In Figure 4.2 the yellow trace is the irradiance profile (1000 → 750 → 400 → 200 W/m² and back to 1000 W/m²), the noisy green trace is the PV output power, and the blue trace is the PV voltage. With a fixed duty cycle the operating point does not follow the MPP: even at 1000 W/m² only about 250–270 W is extracted, compared with about 425 W available. When the irradiance drops to 200 W/m² (at 6 s) the output power falls to almost zero, although about 80–90 W should be available at that irradiance, and the PV voltage falls to about 10 V instead of the 55–60 V expected from the PV characteristic. This shows the need for an MPPT algorithm.

4.3 Power Tracking of P&O Algorithm under Step-Varying Irradiance without PI Controller

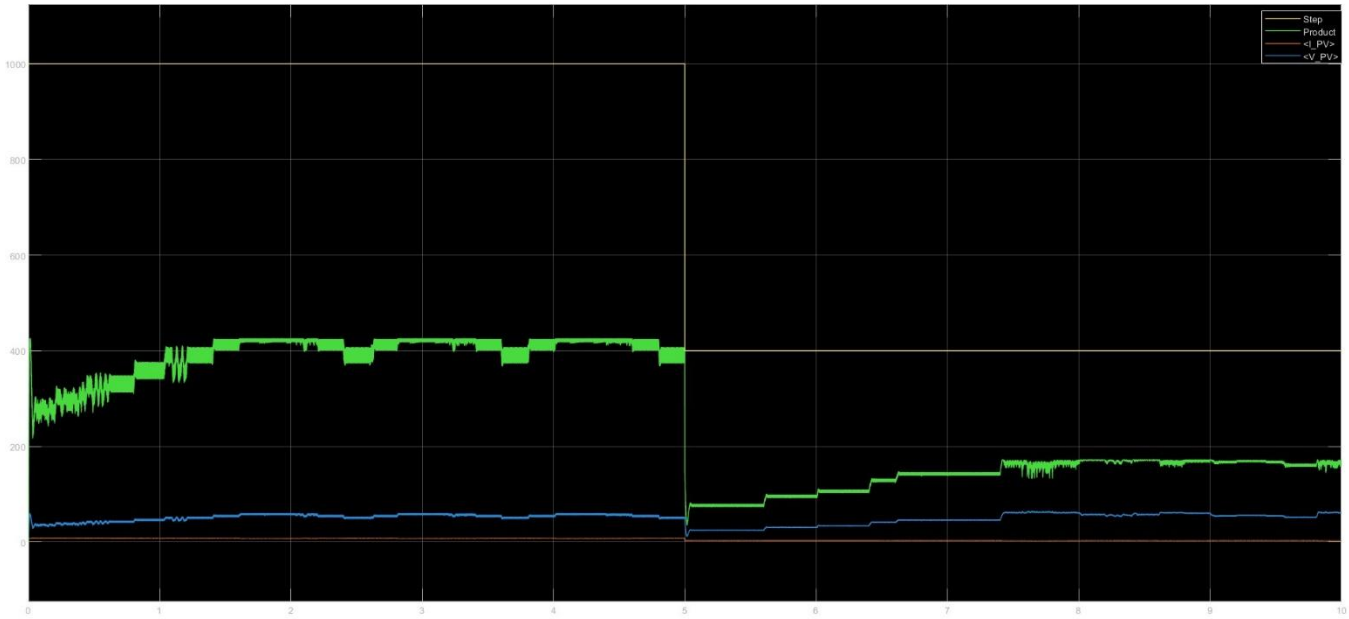


Figure 4.3: P&O operation under step varying irradiance

Figure 4.3 shows P&O without a PI controller under a step change in irradiance from 1000 W/m² to 400 W/m² at 5 s. The power increases stepwise towards the maximum point of about 420 W at 1000 W/m², and the step-based search causes some high-frequency oscillation at steady state, i.e. around the maximum point. Reaching steady state takes quite a long time, about 1.5 s here. This depends on the step size; a step size of 0.01 was used, which is the reason for the slow tracking. When the irradiance drops, the output power drops and the operating point falls away from the MPP. The algorithm then tracks the new MPP again with stepwise increments, reaching just under 200 W (about 170 W) at 400 W/m². After the irradiance drop it takes even longer to reach the MPP, around 2.5 s.

4.4 MPPT using P&O with PI Controller under Step-Varying Irradiance

Here P&O is run with a PI controller for different values of Kp and Ki, and the values are chosen by trial and error. A large perturbation step of 10 V is applied to the voltage reference to obtain a faster MPPT response; this compromises the smoothness of the search, which the PI controller then helps to restore. The three gain sets compared are shown below.

- For Kp = 1 and Ki = 1

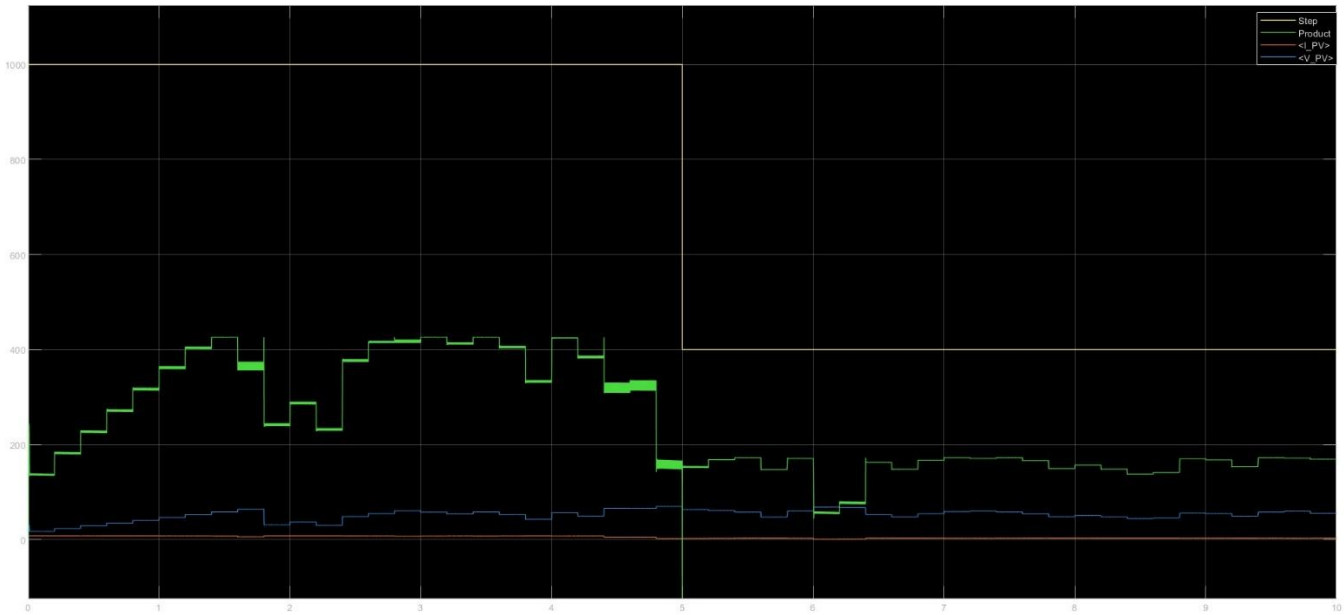


Figure 4.4: P&O operation with Kp = 1 and Ki = 1

- Kp = 0.1 and Ki = 5

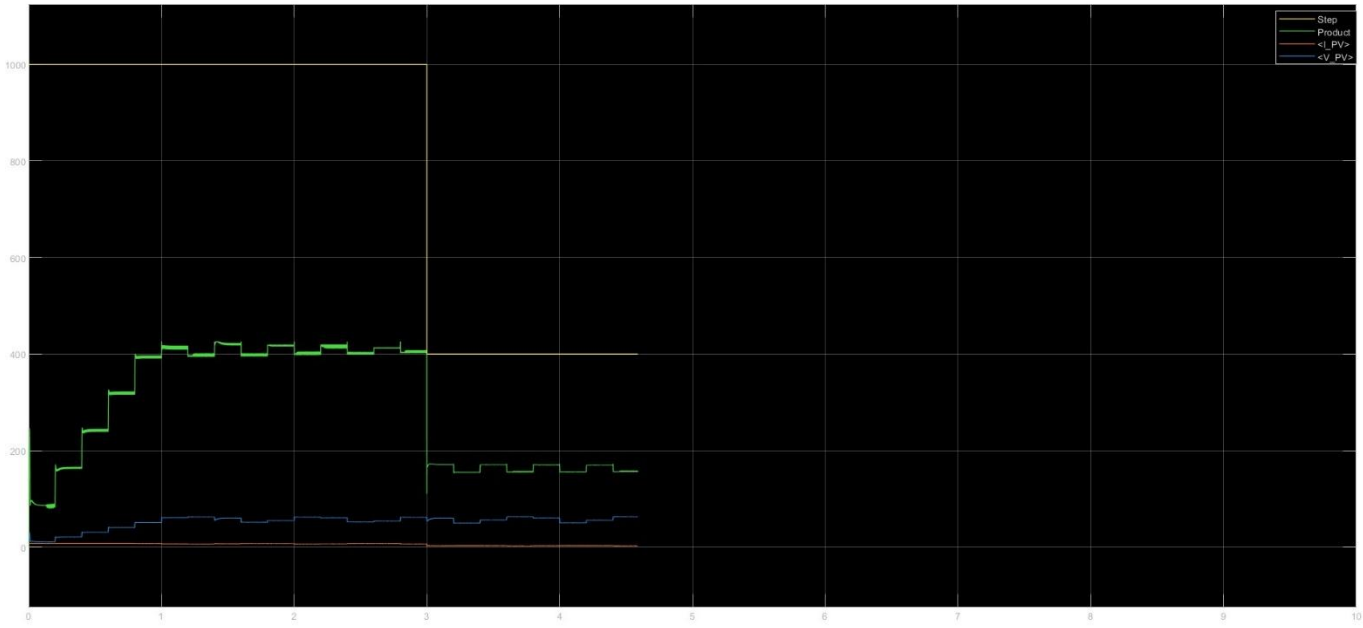


Figure 4.5: P&O operation with Kp = 0.1 and Ki = 5

- Kp = 0.2 and Ki = 20

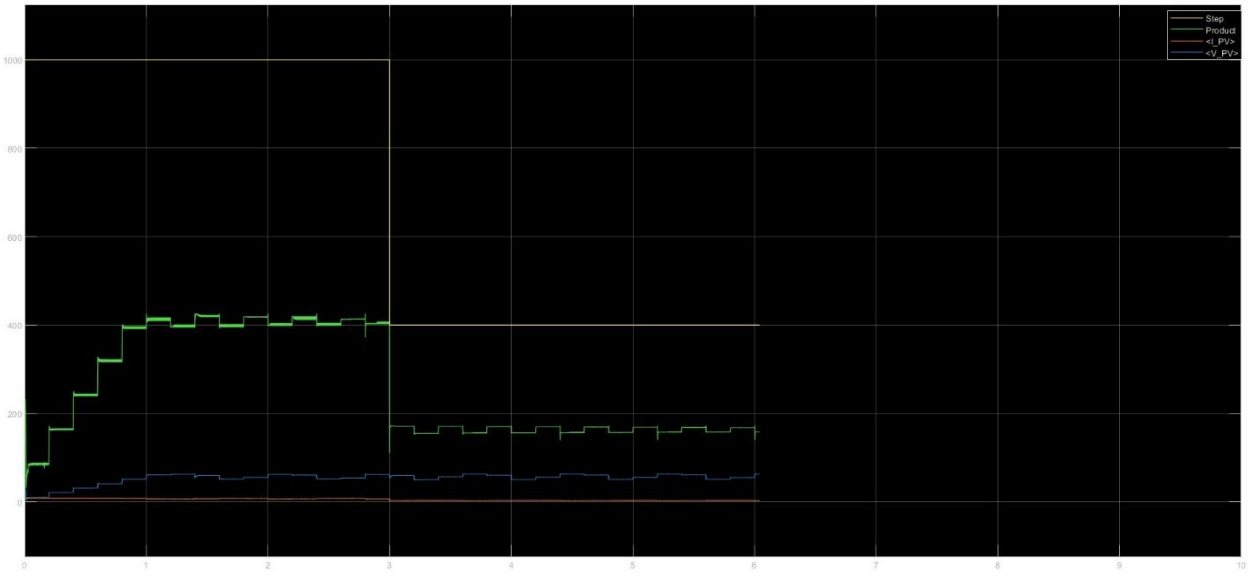


Figure 4.6: P&O operation with Kp = 0.2 and Ki = 20

With Kp = 1 and Ki = 1 (Figure 4.4) the response is irregular: the power repeatedly drops far below the MPP (for example near 1.8 s and 6 s) and does not settle. With Kp = 0.1 and Ki = 5 (Figure 4.5) and with Kp = 0.2 and Ki = 20 (Figure 4.6) the power reaches the MPP of about 420 W in roughly 1 s, recovers quickly after the irradiance step to 400 W/m², and shows only small, regular ripple. Thus the tuned gains reduce the wide oscillations and remove the steady-state error. Note that the irradiance step occurs at 5 s in Figure 4.4 but at 3 s in Figures 4.5 and 4.6.

4.5 MPPT using PSO with PI Controller under Step-Varying Irradiance

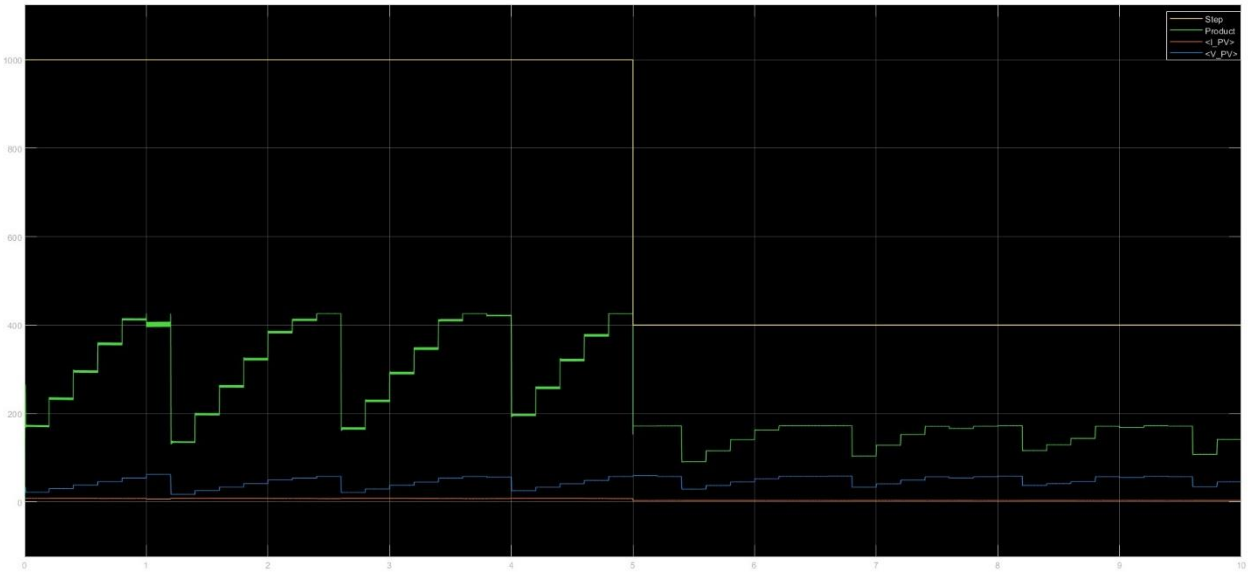


Figure 4.7: PSO operation under step varying irradiance

Figure 4.7 shows very fast tracking of the MPP by PSO: the power reaches about 420 W within roughly 1 s. However, because the swarm keeps exploring the voltage range, the power periodically drops (to about 130–200 W) before climbing back to the MPP, which gives a saw-tooth power curve and wastes energy under uniform irradiance. Since PSO is a global search algorithm, it is expected to perform better on multi-peak curves, i.e. under partial shading.

4.6 Partial Shading Condition

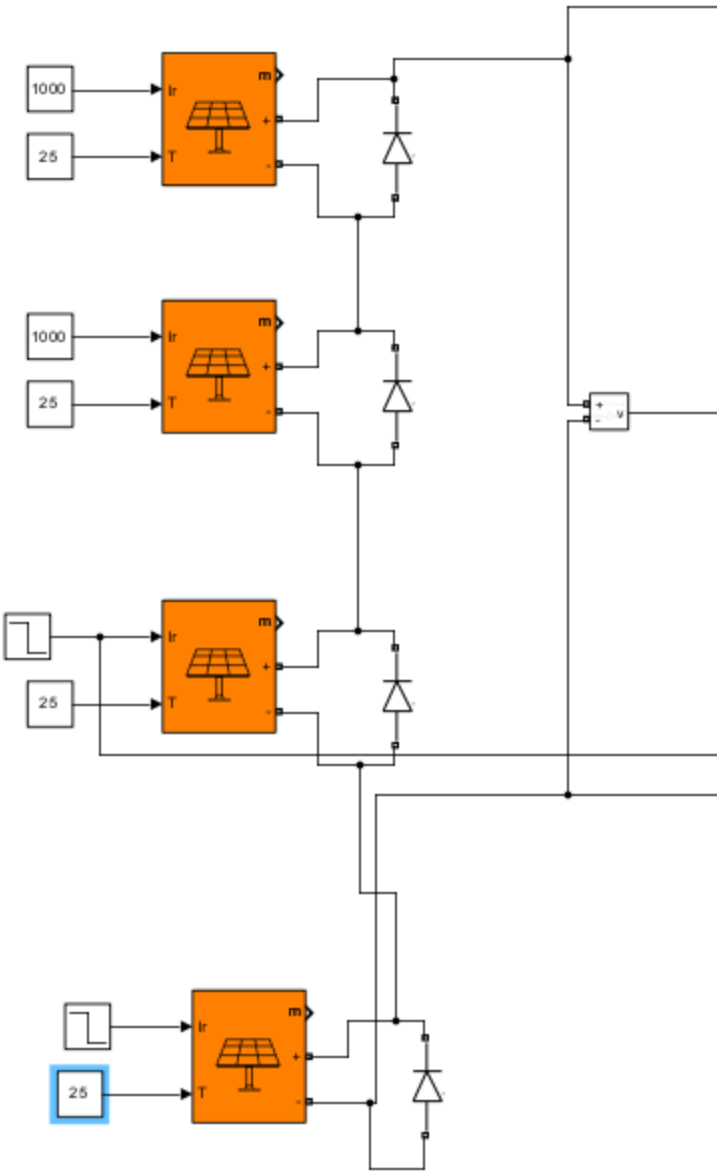


Figure 4.8: Partial Shading configuration in SIMULINK

Four PV Array blocks are connected in series, each through a bypass diode (Figure 4.8). Two arrays receive a constant irradiance of 1000 W/m². The other two receive 1000 W/m² up to 0.5 s, so that the whole string operates as a normal PV array, and then step down to 400 W/m², which causes the partial shading condition.

Figure 4.9 shows the power output of fixed-duty-cycle operation under the partial shading condition.

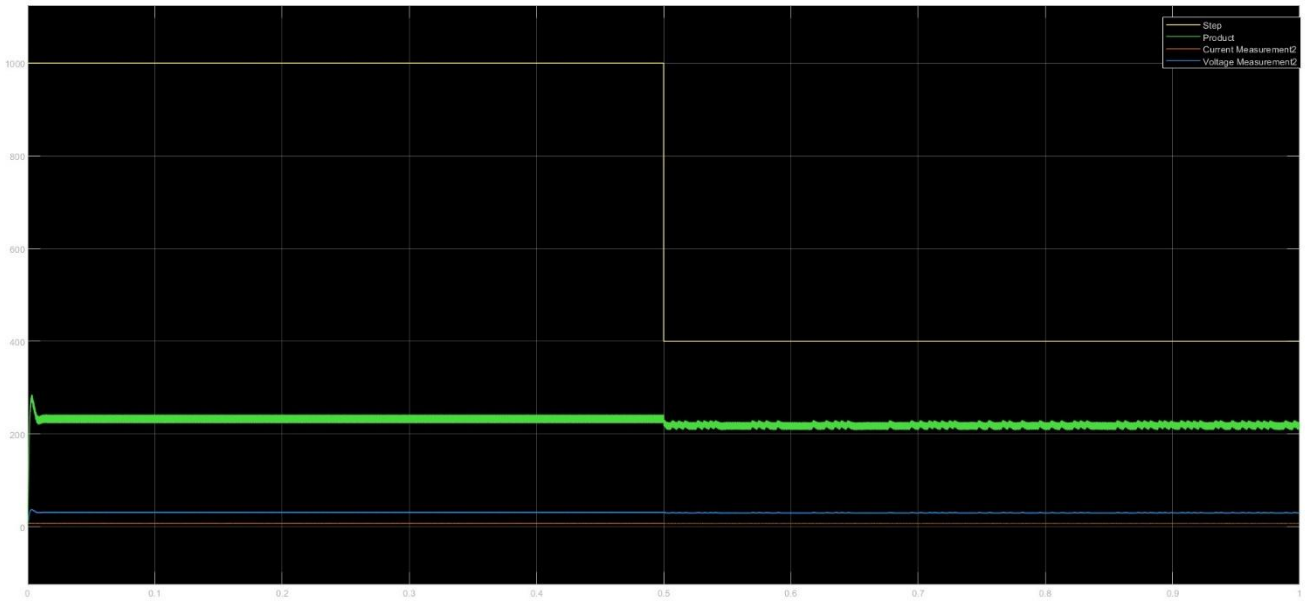


Figure 4.9: Fixed duty cycle operation under partial shading

With a fixed duty cycle the output stays at roughly 230 W before the shading and about 215 W after it, which is far below the power that is actually available (about 485 W before shading, as later found by PSO in Figure 4.12).

4.7 P&O Algorithm Performance under Partial Shading Condition

- Without PI

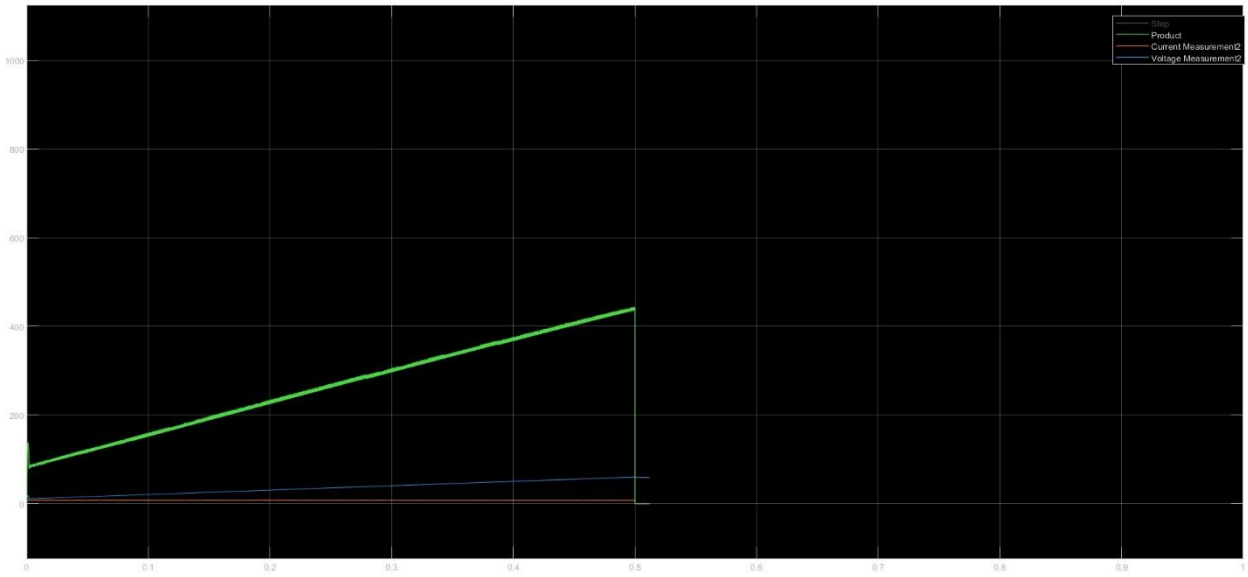


Figure 4.10: P&O under partial shading without PI

- With PI

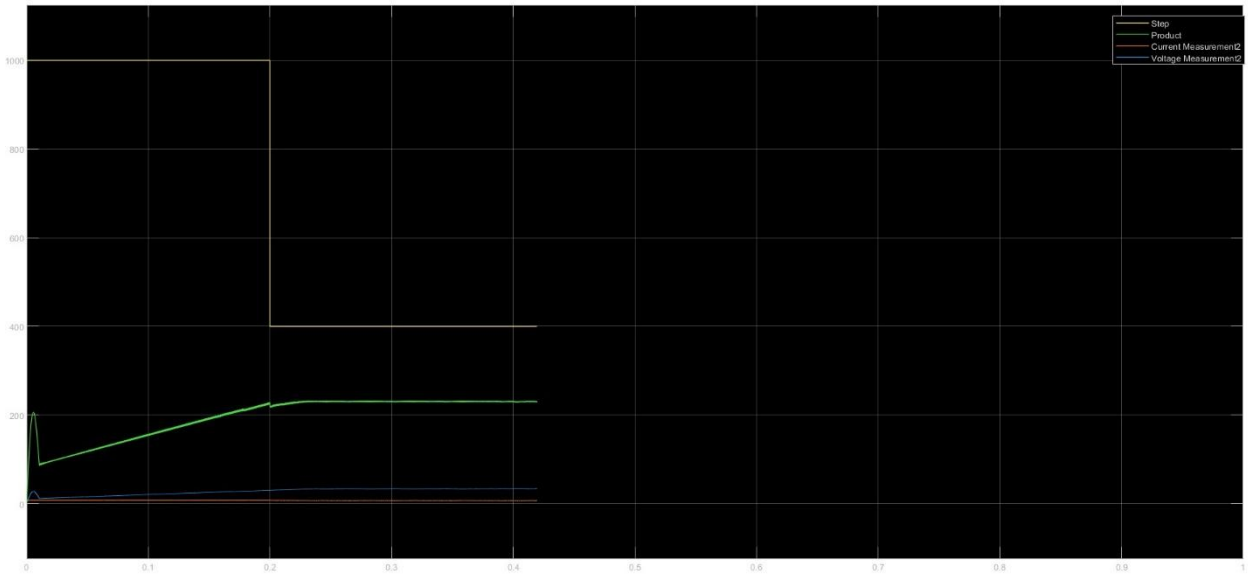


Figure 4.11: P&O under partial shading with PI

Without PI (Figure 4.10) the power rises steadily as the voltage reference is stepped upwards, reaching about 440 W at 0.5 s, but it drops to zero when the shading is applied. With PI (Figure 4.11) the power settles at about 230 W (at roughly 35 V) and remains flat; in this run the shading step occurs at 0.2 s. In both cases P&O stays on the first peak of the multi-peak P-V curve instead of moving to the global one, because it only follows the local gradient.

4.8 PSO Algorithm under Partial Shading Condition

- No PI

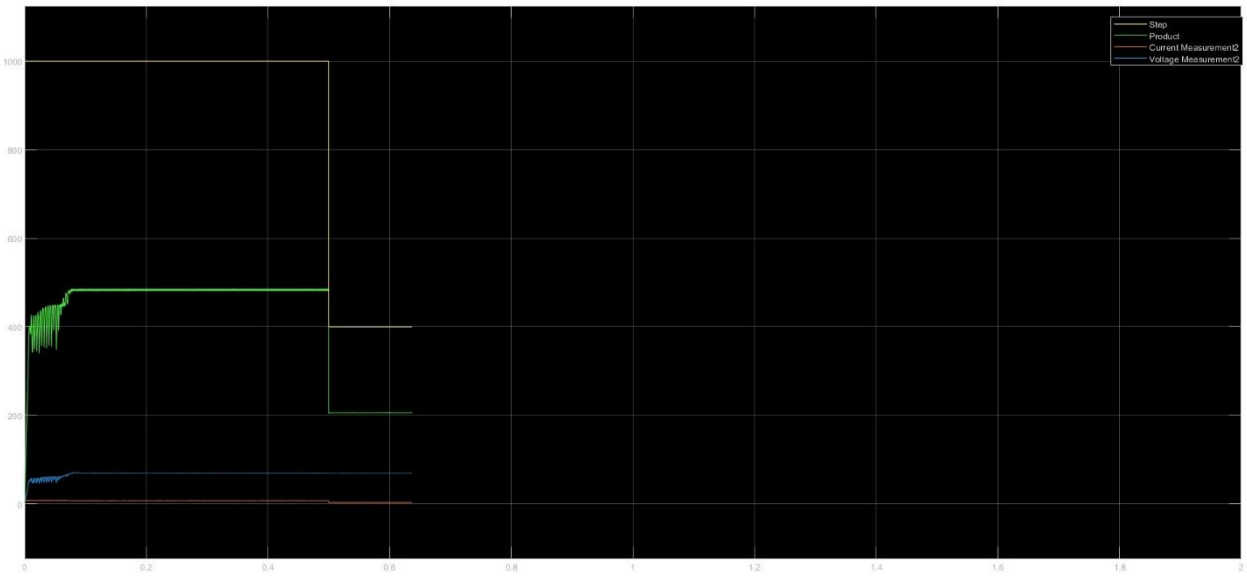


Figure 4.12: PSO under partial shading without PI

- At Kp = 1 and Ki = 1

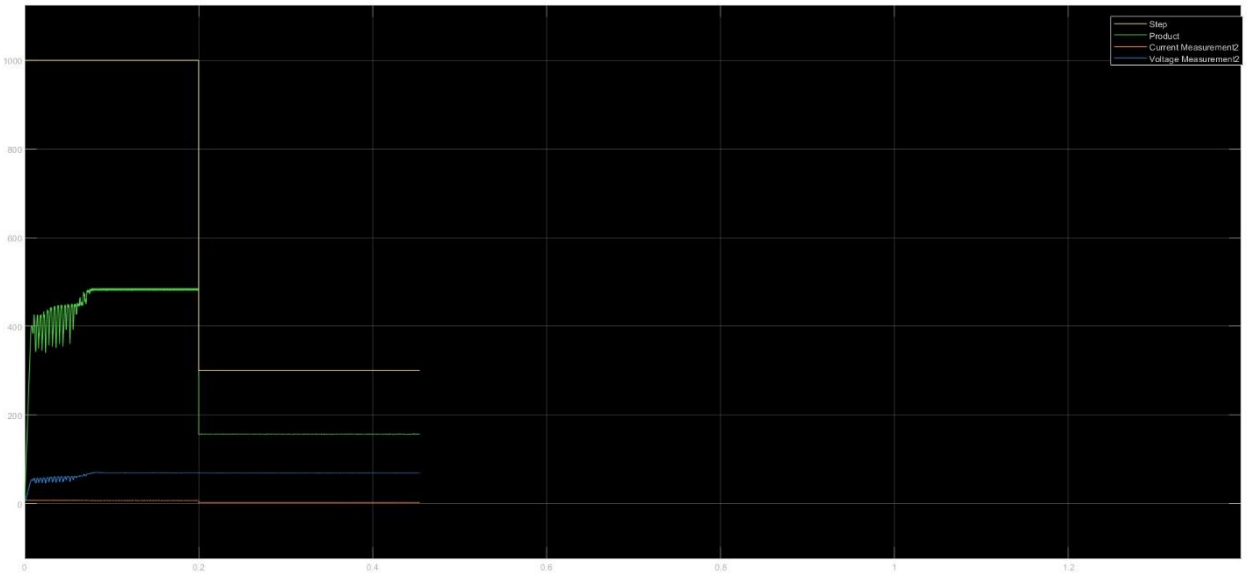


Figure 4.13: PSO under partial shading with Kp = 1 and Ki = 1

- At Kp = 0.1 and Ki = 5

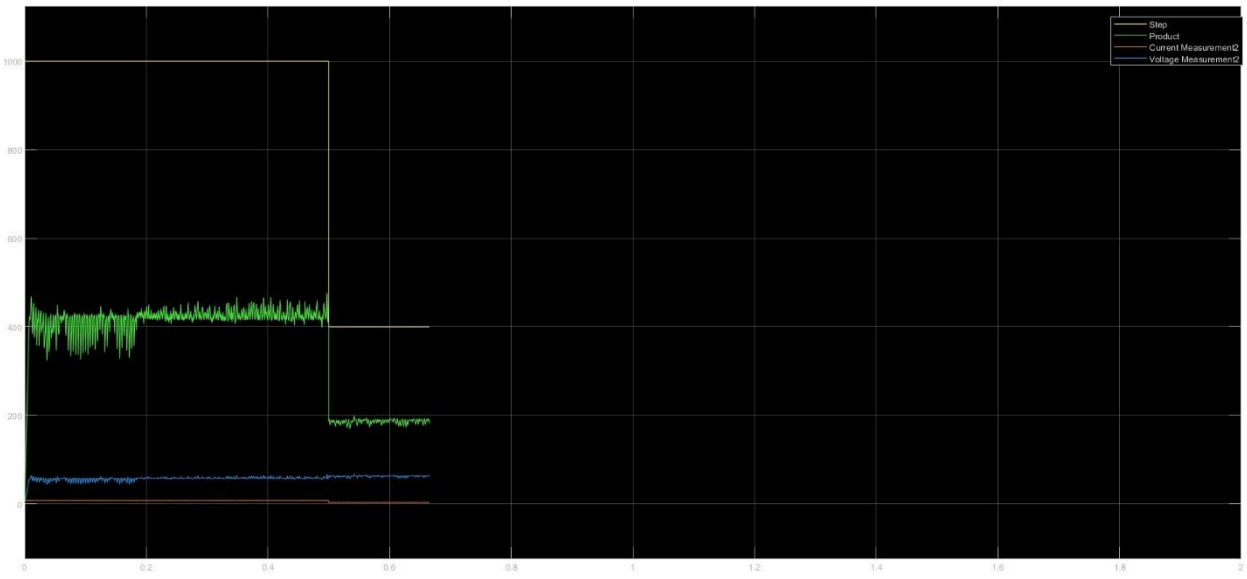


Figure 4.14: PSO under partial shading with Kp = 0.1 and Ki = 5

- At Kp = 0.2 and Ki = 10

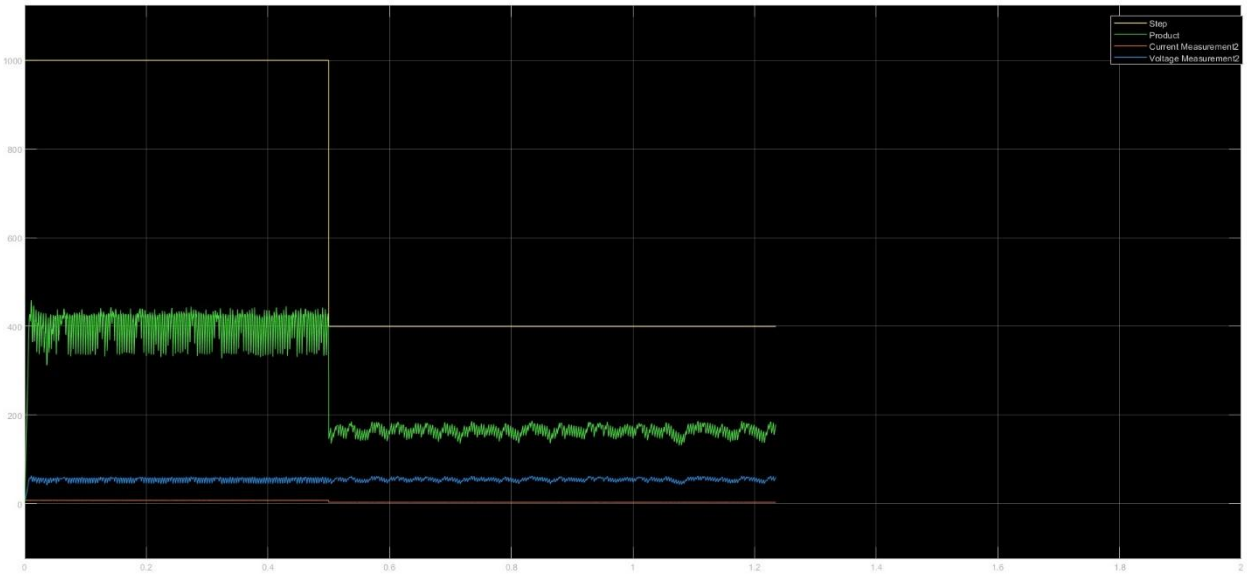


Figure 4.15: PSO under partial shading with Kp = 0.2 and Ki = 10

- At Kp = 0.5 and Ki = 20

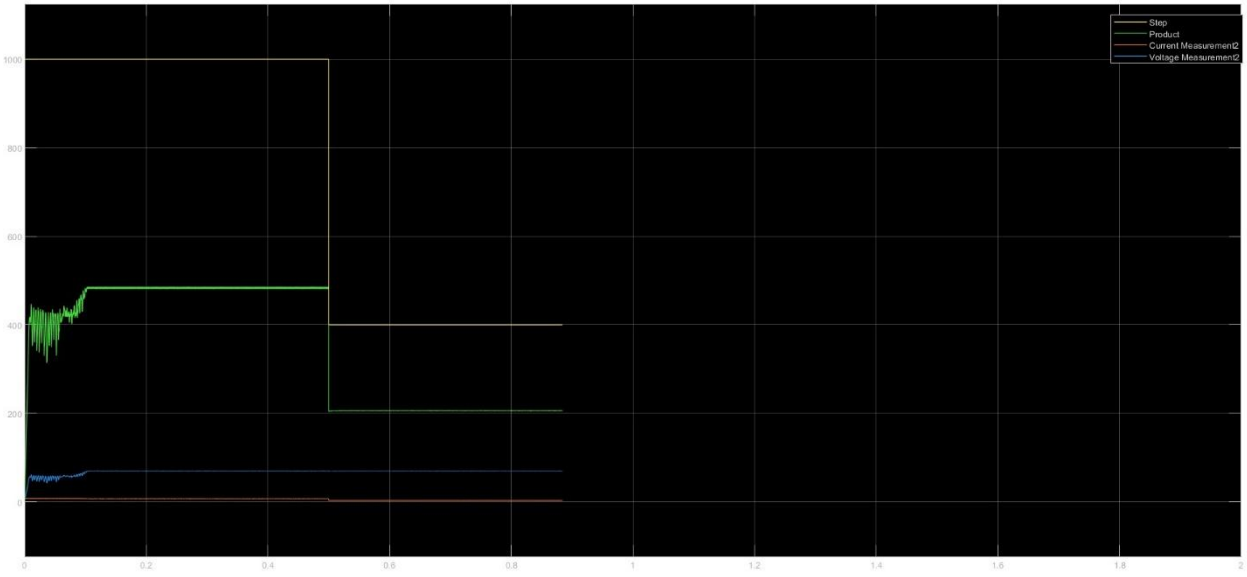


Figure 4.16: PSO under partial shading with Kp = 0.5 and Ki = 20

- At very high Ki, Ki = 1045

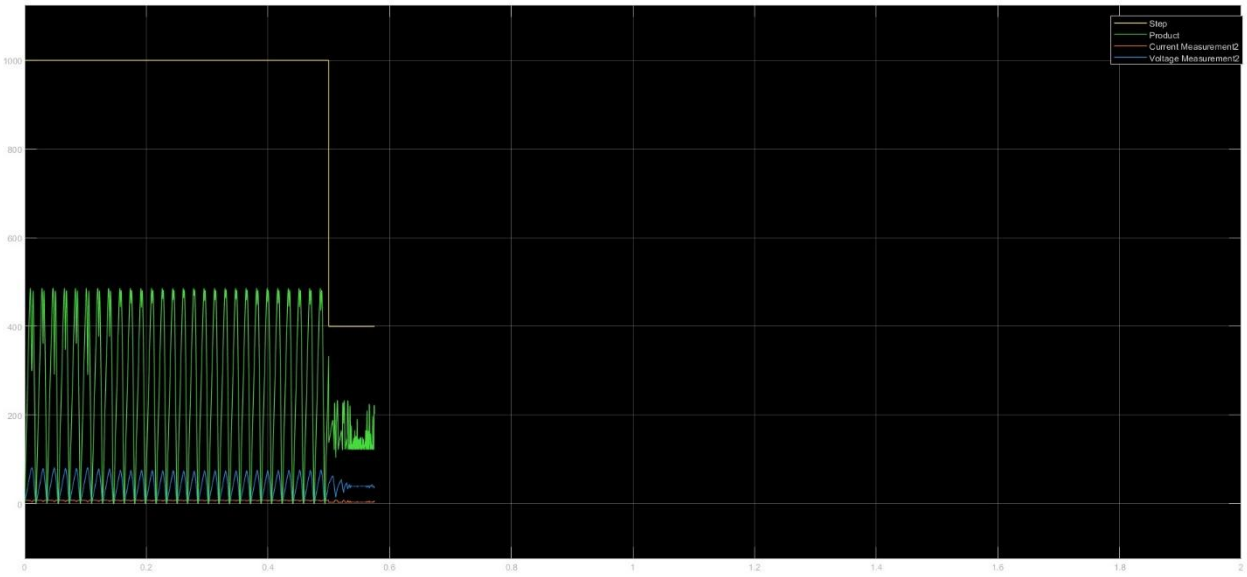


Figure 4.17: PSO under partial shading with very high Ki

In Figures 4.12 and 4.16 the PSO-based tracker converges within about 0.1 s to roughly 485 W at about 70 V and then holds it with negligible ripple, which is much faster than P&O in Section 4.7. The PSO was run for different PI gains (Figures 4.13–4.16), and from trial and error Kp = 0.5 and Ki = 20 showed comparatively the best performance. With an excessively high integral gain (Ki =

1045, Figure 4.17) the loop becomes unstable and the power oscillates continuously, so the integral gain must be limited. After the shading at 0.5 s the power settles at about 205 W and stays constant, i.e. the swarm holds the voltage to which it had converged. Since the controller has no restart on irradiance change, a trigger that re-initializes the swarm after a sudden power change would be needed to guarantee that the new GMPP is tracked after every shading event.

5. CONCLUSION

This project presented a simulation-based study of Maximum Power Point Tracking under Partial Shading Conditions in PV systems. By analyzing the limitations of the conventional PI-tuned P&O algorithm, it shows a clear need for global search techniques. A PSO-based global MPPT, combined with a tuned PI controller for duty cycle generation, was implemented in MATLAB/Simulink and gave a fast, well-controlled response.

Power tracking curves were presented for several scenarios of the P&O and PSO algorithms. Under uniform irradiance, PSO fluctuates because the swarm keeps re-exploring the voltage range, so P&O with a tuned PI controller performs better there. When the same algorithms are subjected to partial shading, the situation is reversed: PSO converges within roughly 0.1 s to a high-power operating point, while P&O, being a local hill-climbing method, can remain stuck on a local peak depending on its starting condition.

Hence the results illustrate why PSO is better suited than P&O under Partial Shading Conditions. A more practical implementation should combine both: PSO brings the operating voltage close to the global peak, after which P&O takes over to reach the top of the peak precisely. Adding a restart trigger on irradiance change and validating the scheme on hardware are the natural next steps.

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APPENDIX-I: Code for P&O Algorithm Used in MATLAB Function Block

```
function V_ref = fcn(V, V_prev, P, P_prev, V_ref_in)

  step = 10;

  if(P>P_prev)
    if(V>V_prev)
      V_ref = V_ref_in + step;
    else
      V_ref = V_ref_in - step;
    end
  else
    if(V>V_prev)
      V_ref = V_ref_in - step;
    else
      V_ref = V_ref_in + step;
    end
  end

  if(V_ref>70)
    V_ref = 70;
  end
  if(V_ref<10)
    V_ref = 10;
  end
```

APPENDIX-II: Code for PSO Algorithm Used in MATLAB Function Block

```
function Vref_out = pso_mppt_vref_controller(V_pv, I_pv)
% PSO MPPT Controller (Outputs Vref)
% Inputs:  V_pv     - Measured PV Voltage (V)
%          I_pv     - Measured PV Current (A)
% Output:  Vref_out - Reference Voltage output (V)

    %% 1. Algorithm Parameters
    N = 4;          % Swarm size
    w = 0.5;        % Inertia weight
    c1 = 1.5;       % Cognitive acceleration coefficient
    c2 = 2.0;       % Social acceleration coefficient

    % Search bounds for PV Voltage (Adjust based on PV module datasheet)
    V_min = 5.0;     % Minimum search voltage (V)
    V_max = 70;    % Maximum search voltage / near Voc (V)
    v_max = 5;     % Maximum step change per iteration (V)

    %% 2. Persistent Variables
    persistent p position velocity pbest_v pbest_p gbest_v gbest_p initialized;

    if isempty(initialized)
        % Initialize particles uniformly across voltage range
        position = V_min + (V_max - V_min) * (1:N)' / (N + 1);
        velocity = zeros(N, 1);
        pbest_v  = position;
        pbest_p  = zeros(N, 1);
        gbest_v  = (V_min + V_max) / 2;
        gbest_p  = -Inf;
        p        = 1;
        initialized = true;
    end

    %% 3. Power Evaluation
    P_pv = V_pv * I_pv;

    %% 4. Update Personal & Global Bests
    if P_pv > pbest_p(p)
        pbest_p(p) = P_pv;
        pbest_v(p) = position(p);

        if P_pv > gbest_p
            gbest_p = P_pv;
            gbest_v = position(p);
        end
    end

    %% 5. Advance Particle / PSO Update
    p = p + 1;

    if p > N
        p = 1; % Reset to particle 1
```

```
        for i = 1:N
            r1 = rand();
            r2 = rand();

            % Velocity & Position update for Vref
            velocity(i) = w * velocity(i) ...
                    + c1 * r1 * (pbest_v(i) - position(i)) ...
                    + c2 * r2 * (gbest_v - position(i));

            velocity(i) = max(min(velocity(i), v_max), -v_max);
            position(i) = position(i) + velocity(i);
            position(i) = max(min(position(i), V_max), V_min);
        end
    end

    %% 6. Output Target Reference Voltage
    Vref_out = position(p);
end
```